

Components of Artificial Neural Network

Neuron / Node / Perceptron

- **Definition:**

The smallest unit of an Artificial Neural Network, inspired by biological brain neurons.

- **Explanation:**

Each neuron takes inputs, multiplies them with weights, adds a bias, applies an activation function, and produces an output.

- **Formula:**

$$y = f(\sum(x_i \cdot w_i) + b)$$

- where:

x_i = input values

w_i = weights

b = bias

f = activation function

- **Example :**

Suppose a student's marks in Maths = 80, Science = 70. The network predicts "Pass/Fail". ◦ Input = [80, 70]

◦ Weights = [0.5, 0.4], Bias = 10

◦ Output = $(80 \times 0.5 + 70 \times 0.4 + 10) = 40 + 28 + 10 = 78 \rightarrow$ Passed.

Input Layer

- **Definition:**

The first layer of the network where raw data enters.

- **Explanation:**

Each node in this layer represents one feature from the dataset.

- **Example:**

In handwriting recognition (MNIST), each image (28×28 pixels) is flattened into **784 inputs**.

Hidden Layer

- **Definition:**

Layers between input and output layers where transformations occur.

- **Explanation:**

Hidden layers learn complex patterns by applying weights, bias, and nonlinear functions.

- **Example:**

In image classification, hidden layers detect **edges, shapes, and higher-level features** step by step.

Output Layer

- **Definition:**

The final layer that produces the network's predictions.

- **Explanation:**

Its size depends on the task:

- Binary classification → 1 node (e.g., Yes/No).
- Multi-class classification → N nodes (softmax gives probabilities).

- **Example:**

Predicting digits 0–9 → Output layer with **10 neurons**.

Weights

- **Definition:**

Numerical values representing the strength of connections between neurons.

- **Explanation:**

- High weight → strong influence of input on output.
- Low/zero weight → weak/no influence.

- **Example:**

In a spam filter, the word "free" may get a **high weight** (strong spam indicator).

Bias

- **Definition:**

An additional parameter that allows shifting the activation function.

- **Explanation:**

It helps the model fit data better, even if all inputs are zero.

- **Example:**

In linear regression y

$$y = mx + c$$

$y=mx+c$, the **bias** is like the intercept c